

ME 149 – Lab 1: Oscilloscope, Function Generator, and Multimeter

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Overview

I initially measured the 9 V battery's open circuit voltage using a digital multimeter. I then measured a 1 kHz sine wave which is 20mV peak to peak using a Tektronix TBS2000 oscilloscope. I found that any modifications to the trigger would cause a shift in the phase of the wave. I was able to plot the waveform data that I collected on MATLAB.

Experimental Setup

1. Battery Voltage Measurement:
 - A digital multimeter was used in DC voltage mode.
 - The leads were placed across the battery terminals, and a reading was recorded
2. Oscilloscope and Function Generator
 - The Tektronix AFG3021 Function Generator was set to output a 1 kHz sine wave with a 20 mV peak-to-peak amplitude
 - A BNC cable connected the generator output to Channel 1 of the Tektronix TBS2000 oscilloscope
 - The oscilloscope was set to AC coupling, internal triggering, and Run mode. Autoset was used to display multiple cycles of the signal
 - The trigger level was adjusted to observe how the wave's phase could be shifted horizontally
 - A single waveform capture was saved in CSV format ("tek00aa.csv") to a USB drive and then plotted in MATLAB

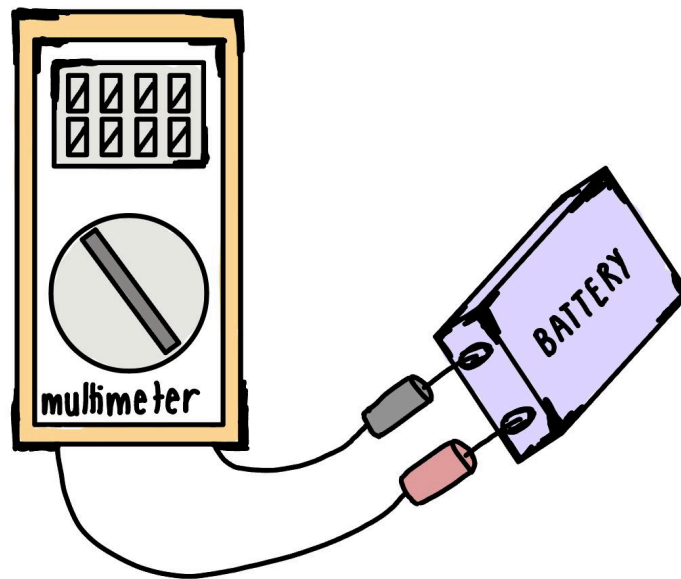


Figure 1: First *Experimental Setup*. Voltage reading was recorded from a multimeter set to DC voltage mode

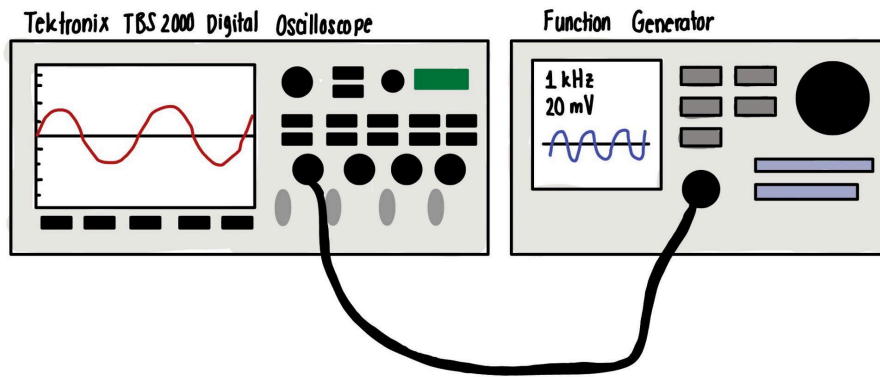


Figure 2. I connected the Tektronix TBS2000 oscilloscope to the Tektronix AFG3021C function generator using a BNC cable. I set the function generator to output a sine wave at 1 kHz with a 20 mV peak-to-peak amplitude. I then used AC coupling on the oscilloscope to display the waveform. I then adjusted the trigger level to shift the phase of the wave horizontally and saved it to a USB drive.

Results

1. **Battery Voltage:** The measured voltage of the 9 V battery was 9.50 V
2. **Sine Wave Frequency:** From the data, the period T was approximately 0.001 s per cycle, giving a frequency of $f = 1/T = 1/0.001\text{ s} = 1000\text{ Hz} = 1\text{ kHz}$
3. **Signal Magnitude:** The measured peak voltages were approximately +10.38 mV and -9.62 mV, summing up to an amplitude of about 20 mV.
4. **Maximum Phase Shift (Trigger):** By adjusting the trigger level, the measured shift in the displayed waveform was about 400 μs total, showing how far in time the wave could be moved from approximately -6 μs to +394 μs around the trigger reference

Voltage vs. Time Plot

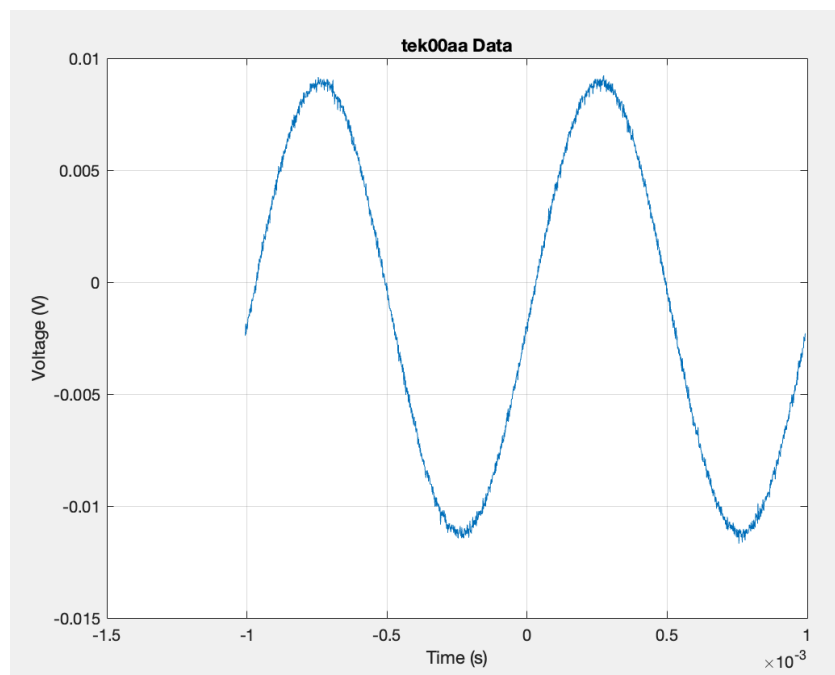


Figure 2. Voltage vs. Time plot captured from the oscilloscope for 1 kHz, 20 mV peak-to-peak sine wave. The data was collected from the Tektronix TBS2000 oscilloscope and saved in CSV format. The time axis shows a period of 1 ms, matching the 1 kHz frequency. The voltage axis shows peaks near $\pm 10\text{ mV}$, consistent with the function generator's 20 mV setting

